

Application: IoT Data Processing Pipeline using Master Theorem

Source Code (C Program):

```
#include <stdio.h>
#include <math.h>

// Function to calculate log base 2
double log2_custom(double n)
{
    return log(n) / log(2);
}

int main()
{
    int n;

    printf("Enter the size of IoT sensor data (n): ");
    scanf("%d", &n);

    printf("\nRecurrence Relation:\n");
    printf("T(n) = 3T(n/3) + nlog(n)\n");

    printf("\nMaster Theorem Parameters:\n");
    printf("a = 3\n");
    printf("b = 3\n");

    double exponent = log(3) / log(3);

    printf("log_b(a) = %.0lf\n", exponent);
```

```
double nlogn = n * log2_custom(n);

printf("\nn log n = %.2lf\n", nlogn);

printf("\nAnalysis:\n");
printf("n^(log_b(a)) = n\n");
printf("f(n) = n log n\n");
printf("Since f(n) = Theta(n log n), it matches Case 2 of Master Theorem.\n");

printf("\nFinal Time Complexity:\n");
printf("T(n) = Theta(n (log n)^2)\n");

return 0;
}
```